

Atomistic Understanding of Ion Exchange Strengthening of Boroaluminosilicate Glasses: Insights from Molecular Dynamics Simulations and QSPR Analysis

Po-Hsuen Kuo and Jincheng Du*



Cite This: *J. Phys. Chem. B* 2022, 126, 2060–2072



Read Online

ACCESS |



Metrics & More

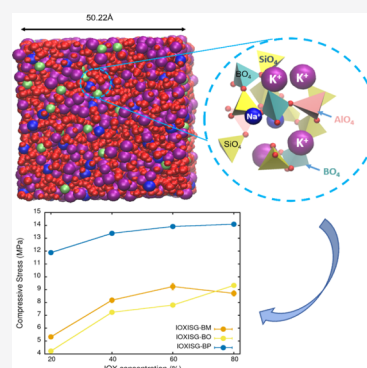


Article Recommendations



Supporting Information

ABSTRACT: Ion exchange (IOX) is an effective and widely used method to enhance mechanical properties of various glass products ranging from the touch screen of consumer electronics to window shields of airplanes and spacecrafts. IOX or chemical strengthening is achieved through the creation of a compressive surface layer on the glass product. Although widely studied experimentally, the fundamental understanding of the IOX strengthening process is still limited. In this work, we have applied large-scale atomistic simulations to understand IOX-induced mechanical property changes and their relation to the glass composition and structural characteristics. Two series of borosilicate glasses are studied to elucidate the composition effect, with boron oxide for silica and alumina for silica substitutions, respectively, on the mechanical properties of different levels of K^+ to Na^+ ion exchanges by using molecular dynamics (MD) simulations with a set of recently developed effective partial charge potentials. The linear network dilation coefficient (LNDC), a common measure of IOX behaviors, was calculated for each of the glass compositions. Quantitative structural property relationship (QSPR) analysis based on the MD-generated structural features was used to establish the structure–property correlations of mechanical and other properties. The results show strong composition dependence of the LNDC, hence the suitability of IOX strengthening. This behavior is discussed based on glass structure features of the glasses. It was found that glass compositions with a higher amount of mixed glass formers, higher network connectivity, and less complex components tend to show higher calculated LNDC and higher surface compressive stress. MD simulations, in combination with QSPR analysis, can thus provide atomistic insights into how the glass composition and structural characteristics affect IOX behaviors.



1. INTRODUCTION

The ion exchange (IOX) process, or chemical strengthening, has been widely used in the glass industry and advanced technologies to enhance the strength and toughness of glass products ever since its first development in early 1960s.¹ Unlike conventional thermal strengthening, IOX can be applied to ultra-thin glass products and has high dimensional tolerance, thus leading to numerous applications ranging from touch screens in consumer electronics, aerospace and high-speed train windshields, ophthalmic glass, and substrates of hard drives for information storage and electronic circuits.^{2–5} IOX, also known as IOX stuffing, chemical tempering, or chemical strengthening, is a surface modification technique that involves thermally activated interdiffusion of alkali ions between the glass and the molten salt bath. Larger cations (such as K^+) from the molten salt diffuse into the surface layer of the glass, exchanging with smaller cations (such as Na^+) from that glass that diffuse out to the molten salt, creating a compressive surface layer that improves the mechanical strength of the glass product. The temperature of the IOX process is usually kept below the glass transition temperature (T_g) of the glass to reduce stress relaxation that can reduce the

strengthening effect. Typically, IOX can generate a compressive surface layer of tens to hundreds of microns thick, leading to several hundreds of megapascals of surface compression.⁶ This surface compressive stress layer can prevent crack initiation and propagation when encountering external impacts, hence improving the strength and damage resistance of the glass products. Despite widespread industrial applications, the exact mechanism of the IOX strengthening is still not fully understood. This is partly due to the lack of suitable in situ characterization methods to probe the structural changes and stress developments during the IOX process. This lack of understanding makes it challenging to rationally design new glass compositions for IOX processing and applications. Atomistic simulation methods such as molecular dynamics (MD) have already been widely adopted to study glass

Received: December 29, 2021

Revised: February 9, 2022

Published: February 24, 2022



structures. These methods can potentially address the challenges and provide insights into the structural changes at the atomic scale during the IOX processes. Recently, the simulation- and modeling-based material genome approaches have been used to accelerate the development of various types of materials. In this study, we aim to combine physics-based atomistic simulations and data-driven techniques such as quantitative structural property relationship (QSPR) analysis as a new approach to understand the IOX processes and associated mechanisms with a goal to gain insights into how the glass composition and structures affect the effectiveness of the IOX processes.

The mathematical expression of the stress profile generated by IOX is given as eq 1

$$\sigma(z) = -\frac{BE}{1-\nu}[C(z) - C_{\text{avg}}] \quad (1)$$

where $\sigma(z)$ is the compressive stress at the penetration depth z reached by the alkali ion during the IOX process, E is Young's modulus of the glass, and ν is Poisson's ratio. It is worth noting that eq 1 does not take stress relaxation into consideration. For the force balance condition, the average concentration of substituting alkali ions, C_{avg} , is subtracted from the local concentration of substituting alkali ions, $C(z)$, allowing the net stress to be zero; that is, the integration of compressive stress from the surface to the maximum penetration depth is compensated by the tensile stress, therefore resulting in a zero internal stress. Parameter B is the linear network dilation coefficient (LNDC), which is also known as the "Cooper coefficient".^{7,8} The LNDC gives the linear strain per unit change of alkali ion concentration, which can be expressed as

$$B = \frac{1}{3} \left(\frac{1}{V} \frac{\partial V}{\partial C} \right) \quad (2)$$

where the factor $1/3$ is converting the volume to a linear coefficient of thermal expansion for isotropic materials, V is the volume of the glass, and C is the concentration of exchange ions in IOX. By combining eqs 1 and 2 above, $-E/1-\nu$ can be considered as a factor that converts B , the strain caused by IOX (LNDC), to the stress induced by IOX in the glass. By ignoring the stress relaxation contribution, the compressive stress is directly proportional to the strain induced by per unit of IOX concentration.

In the early studies, it was found that the value of the LNDC (B) calculated from eq 2 is 2–4 times higher than that derived from eq 1 (with the values of $\sigma(z)$, $C(z)$, E , and ν obtained from experimental measurements). For example, in $15\text{Na}_2\text{O}-10\text{CaO}-75\text{SiO}_2$ glass with $\text{Na}^+ \leftrightarrow \text{K}^+$ IOX, the calculated $B \approx 1.3 \times 10^{-3}$ ([mol % K_2O^{-1}]), whereas the experimentally measured value of $B \approx 3-6 \times 10^{-4}$ ([mol % K_2O^{-1}]). This discrepancy between experimentally measured and theoretically calculated B values results in a lower stress profile in IOX glasses than what the model predicted. Because the IOX temperature is significantly lower than T_g , the discrepancy cannot be easily explained by the viscous relaxation of glass.^{4,7-14} By adopting MD simulations, it has been found that this network dilation discrepancy originates from the short- and medium-range atomic structural changes of glass during IOX. Varshneya et al.,^{4,9,10,15,16} Mauro et al.,^{12,14,17} and Tandia et al.^{18,19} found that the anomaly is mainly due to the structural rearrangement after IOX. Tandia et al. showed that the change of the local environment of cations affects the value of the LNDC. During the $\text{Na}^+ \leftrightarrow \text{K}^+$ IOX process, K^+ quickly

moves into a Na^+ site but without a commensurate molar volume change as compared to that of the compositionally equivalent as-melted (CEAM) K^+ -stuffed glass. The volume difference between the two glasses generated from the IOX process and the melt-and-quench process creates an elastic strain that introduces a stress within the glass of the same composition. Tandia et al. described the IOX glass and the CEAM glass as two different glasses with the same composition but acquired through two different types of relaxation processes.^{18,20} After salt bath immersion, the volume of IOX glass expands elastically via the fast " β -relaxation", allowing larger ions to be accommodated into the glass structure without altering the major Si–O network in the glass structure. The volume of CEAM glass expands much slower via the " α -relaxation", which requires higher temperature for glass network viscous flow.^{4,11,12,15,18} The results show a strong connection between the mechanical and structural properties of IOX glasses and the local environment change of major alkali ions, K^+ and Na^+ . As most of the investigations were carried out in silicate glasses or aluminosilicate glasses, such as $\text{Na}_2\text{O}-\text{K}_2\text{O}-\text{SiO}_2$, $\text{Na}_2\text{O}-\text{K}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2$, or $\text{Na}_2\text{O}-\text{CaO}-\text{SiO}_2$, it is worth exploring more complicated compositions with multiple glass formers, especially those with B_2O_3 . By using MD simulation, this paper investigates the effectiveness of mechanical property enhancement by IOX in two series of borosilicate and aluminoborosilicate glasses.

QSPR analysis is an approach to build the correlation between the predetermined structural descriptors of a material and its properties. By establishing the statistical relationship between one or multiple predefined descriptors and the targeted properties of the material, the mathematical relationship can be established.²¹ The mathematical relationship can be considered as a guidance for the inverse design of the material composition or processing method to achieve the desired properties predicted by the QSPR analysis.²¹⁻²⁴ Combing the vast amount of research data accumulated throughout the history and the thriving data analysis method in recent years, QSPR and other related machine learning analysis has become increasingly important in glass material design as compared to the conventional trial-and-error exploration approach of material development. For example, using different descriptors, QSPR analysis has been proven to be successful in predicting the properties of polymer material, such as glass transition temperature (T_g),²⁵⁻²⁶ bioactivity,^{22,27} and Young's modulus.²⁸ In the inorganic glass field, Lusvardi et al. discovered that by using the theoretical structural descriptor (F_{net}), QSPR analysis can also be used to predict the properties of silica-based and phosphor-silicate-based bioactive glass, such as crystallization temperature (T_c), T_g , and bioactivity.²⁹⁻³² Our recent studies expanded the feasibility of QSPR analysis-based structural descriptor (MD simulation calculated F_{net}) from MD simulations of multicomponent glasses with mixed network formers, such as SiO_2 , Al_2O_3 , ZrO_2 , and B_2O_3 . The result shows that the F_{net} gives an excellent linear correlation with properties such as the density, T_g , and the dissolution rate (r_0) from experiments.³³⁻³⁵ The good agreement indicates the great potential of QSPR for compositional design of complex multicomponent glasses. Therefore, the correlation of mechanical properties and composition of glass can be established using QSPR, and the optimal condition can be obtained.

In this work, we chose two series of multicomponent boroaluminosilicate glasses, named NQ and international

simplified glass (ISG) series, respectively, with the SiO_2 being gradually substituted by $\text{Al}_2\text{O}_3 + \text{Na}_2\text{O}$ and $\text{B}_2\text{O}_3 + \text{Na}_2\text{O}$. The six-component ISG glass, developed as a model system to study the chemical durability in the field of nuclear waste disposal, was chosen due to the fact that it has been widely investigated and the detailed structural and property information such as mechanical and chemical durability are available. The NQ series are basically sodium borosilicate glasses with the addition of alumina. These two glass series provide the basis for comparison of more complex versus simpler compositions, as well as the effect of glass formers mixing as a result of the substitutions on the glass structures and IOX efficiency. After the regular melt-and-quench generation of the bulk glass structures in each of the glass compositions, 20, 40, 60, and 80% of Na^+ ions were substituted with K^+ ions as a simulation of the IOX process. Besides the compositional change, the glass was prepared using two methods in MD simulation: (1) annealing (labeled as “-an”), to simulate the typical experimental condition of IOX, and (2) melting-and-quenching (labeled as “-mq”), to simulate CEAM glass. The volume change was determined after equilibrium was reached in MD simulations, and then Young's modulus and other mechanical properties of glasses were calculated using the finite difference method with microstrain process introduced. The gradual increase of IOX ion concentration will be used to represent the glass compositions from the bulk to surface of the IOX process. The structural descriptor F_{net} values of both melt-and-quench and IOX glasses were calculated based on the structural analysis from MD and parameters from previous studies.³³ The influences of glass former substitution and IOX concentration on the mechanical properties and on the structural energy were studied using the QSPR analysis. To the best of our knowledge, there is no similar study on IOX analysis of multicomponent borosilicate glasses. The results provide atomistic insights into the IOX process, especially for more complex borosilicate glass compositions, and show the potential of MD simulation-based QSPR analysis in understanding IOX processes in glass materials.

2. METHODOLOGIES

The compositions of two series of glass used in this work, which are named ISG and NQ, are shown in Tables 1 and 2,

Table 1. Compositions of ISG Series Glass in Molar Percentage

sample	SiO_2	B_2O_3	Na_2O	Al_2O_3	CaO	ZrO_2
ISG-BM	65.2	13.2	10.4	3.8	5.7	1.7
ISG-BO	60.2	16.0	12.6	3.8	5.7	1.7
ISG-BP	55.2	18.8	14.8	3.8	5.7	1.7

Table 2. Compositions of NQ Series Glass in Molar Percentage

sample	SiO_2	B_2O_3	Na_2O	Al_2O_3
NQ-ALM	73.4	12.3	14.3	0.0
NQ-AIO	65.2	12.3	18.4	4.1
NQ-AIP	57.0	12.3	22.5	8.2

respectively. For the ISG series, 5% of SiO_2 in glass was substituted by $\text{B}_2\text{O}_3 + \text{Na}_2\text{O}$. The stoichiometric description of ISG series compositions is $(60.2 + 5x)\text{SiO}_2 - (16.0 -$

$2.8x)\text{B}_2\text{O}_3 - (12.6 - 2.2x)\text{Na}_2\text{O} - 3.8\text{Al}_2\text{O}_3 - 5.7\text{CaO} - 1.7\text{ZrO}_2$ ($x = 1, 0, -1$). $x = 1, 0$, and -1 corresponds to ISG-BM, ISG-BO, and ISG-BP, respectively. It is worth mentioning that ISG-BO glass is the nominal compositions of ISG, which has been deeply investigated since 2013 for its representative characteristics of nuclear waste glass.⁷ As shown in Table 2, 8.2% of SiO_2 was substituted by $\text{Al}_2\text{O}_3 + \text{Na}_2\text{O}$. The stoichiometric description of NQ series glass is $(65.2 + 8.2x)\text{SiO}_2 - 12.3\text{B}_2\text{O}_3 - (18.4 - 4.1x)\text{Na}_2\text{O} - (4.1 - 4.1x)\text{Al}_2\text{O}_3$ ($x = 1, 0$, and -1). $x = 1, 0$, and -1 corresponds to NQ-AIO, NQ-AIP, and NQ-ALM, respectively. The samples are named based on {glass series name}–{substituent glass former}{“O”, “M” or “P”}. The “O”, “M”, and “P” represent “original glass ($x = 0$)”, “minus substituent ($x = 1$)”, and “plus substituent ($x = -1$)”.

2.1. Simulation Details for ISG and NQ Series Glasses.

MD simulation in this study was performed using the large-scale atomic/molecular massively parallel simulator (LAMMPS)³⁶ with partial effective charge pairwise empirical Buckingham potential.

$$V(r_{ij}) = \frac{q_i q_j e^2}{4\pi\epsilon_0 r_{ij}} + A_{ij} \exp\left(\frac{-r_{ij}}{\rho_{ij}}\right) - \frac{C_{ij}}{r_{ij}^6} \quad (3)$$

where V and r_{ij} are the potential energy and interatomic distance between ions i and j , respectively. The first term describes the long-range Coulombic interaction, q_i and q_j are the partial atomic charges of ions i and j , respectively, and ϵ_0 is the electrical permittivity in vacuum. The second and third terms describe the short range interaction in the Buckingham form with the exponent repulsion and the van der Waals attraction. The potential parameters, A_{ij} , C_{ij} , and ρ_{ij} used in this study are listed in Table 3, and the details of the potential

Table 3. Atomic Charges and Buckingham Potential Parameters³⁹

atom pair ($i-j$)	A_{ij} (eV)	ρ_{ij} (Å)	C_{ij} (eV Å ⁶)
$\text{Si}^{2.4}-\text{O}^{-1.2}$	13,702.905	0.193817	54.681
$\text{B}^{1.8}-\text{O}^{-1.2}$	14,550.972	0.171281	28.500
$\text{Na}^{0.6}-\text{O}^{-1.2}$	4383.756	0.243838	30.700
$\text{Al}^{1.8}-\text{O}^{-1.2}$	12,201.417	0.195628	31.997
$\text{Ca}^{1.2}-\text{O}^{-1.2}$	7747.183	0.252623	93.109
$\text{Zr}^{2.4}-\text{O}^{-1.2}$	17,943.394	0.226627	127.650
$\text{K}^{0.6}-\text{O}^{-1.2}$	20,526.972	0.233708	51.489
$\text{O}^{-1.2}-\text{O}^{-1.2}$	2029.220	0.343645	192.580

development can be found in the studies by Deng and Du.^{37,38} The reason to choose this set of potential is due to its capability to reproduce the boron coordination change in wide composition ranges and the availability of compatible potential parameters for all oxides in the two glass series including those for ZrO_2 and K_2O . Each simulation cell contains approximately 10,000 atoms, based on the different compositions, and three parallel tests were carried out for each of the six glass compositions. The initial atomic configurations of ISG and NQ series glasses were generated by randomly placing atoms in a cubic cell. Then, the simulation cell was melted at 6000 K using a canonical ensemble (NVT) for 200 ps, quenched at a rate of 5 K/ps using NVT, and then equilibrated at 300 K using the isothermal–isobaric ensemble (NPT), NVT, and micro-canonical ensemble (NVE) for 100 ps in order. The last configuration of the NVE equilibrium was used as the as-obtained glass.

2.2. MD Simulation of Ion-Exchanged Glasses. The mechanical properties of ISG and NQ series glass as a function of IOX concentration were investigated. Ion-exchanged glasses were obtained by randomly substituting 20, 40, 60, and 80% of Na^+ for K^+ from the base glass obtained from the regular melt-and-quench process followed by thermal annealing. Another way to obtain the glass is from fully melt-and-quenching with the ion-exchanged glass compositions. These glasses were named IOXISG and IOXNQ glasses, followed by -an for the annealing method and -mq for the melt-and-quench method, respectively. For the annealing method, the ion-exchanged glasses were annealed at 700 K using *NVT* for 200 ps, which corresponds to the typical temperature of IOX salt bath in the lab. The glasses were then quenched at a rate of 5 K/ps using *NVT* and equilibrated at 300 K using *NPT*, *NVT*, and *NVE* for 100 ps in each ensemble. The last configuration from *NVE* equilibrium was used as the synthesized glass and named IOXISG-an and IOXNQ-an glasses, respectively. For the melt-and-quench method, the IOX glass composition is heated to a high temperature to achieve full structural relaxation, as described in the previous section. The glass was melted at 6000 K for 200 ps, quenched at a rate of 5 K/ps using *NVT*, and then equilibrated at 300 K using *NPT*, *NVE*, and *NVE* for 100 ps in order. The final configurations of *NVE* equilibrium were used as IOXISG-mq and IOXNQ-mq glasses, respectively. The -mq and -an glasses are used to calculate the LNDC (Figure 1).

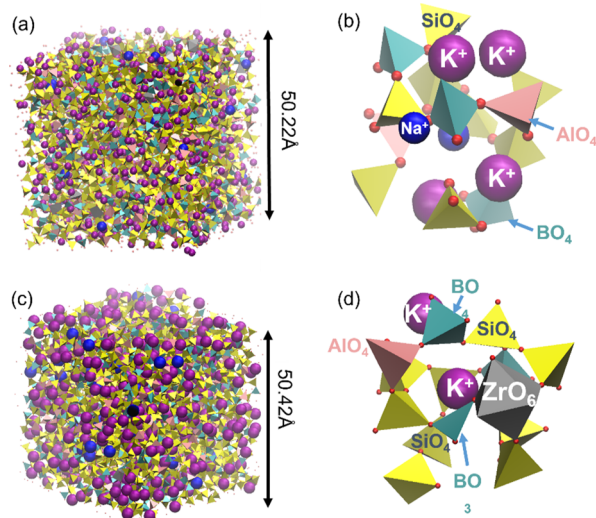


Figure 1. (a) Simulation cell and (b) cation local environment of IOXISG-BM-an with 80% IOX. (c) Simulation cell and (d) cation local environment of IOXISG-BM-mq with 80% IOX.

2.3. Mechanical Property Calculations. Mechanical properties, including Young's, bulk, and shear moduli of ISG, NQ, IOXISG, and IOXNQ series glasses were calculated from the finite difference method with microstrain-induced energy change based on energy minimization using a script within LAMMPS developed by Aidan Thompson⁴⁰ (Figure 2). The method was developed for crystalline materials but has been proven to be feasible in the glass systems in combination with Buckingham potential in our previous study.⁴¹ It is assumed that the system is isotropic, although minute values of off-diagonal stress components might appear due to the finite size

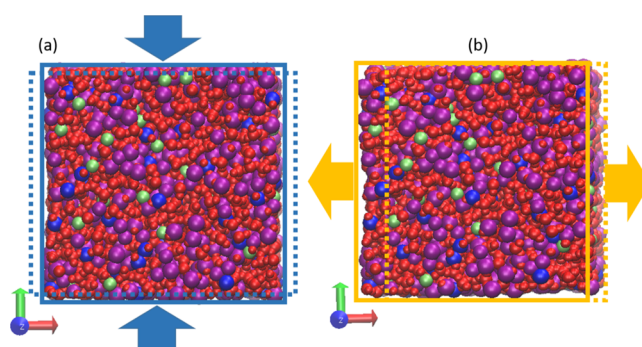


Figure 2. Snapshot of the simulation cell and schematics of strain-induced energy change during mechanical property calculation: (a) compressive strain along the *y*-axis and (b) shear strain parallel to the *x*-*y* plane.

of the simulation cell. These values, however, are relatively small. The elastic stiffness tensor can be given by eq 4

$$\sigma_i = C_{ij} \times \varepsilon_j \quad (4)$$

where σ_i is the stress tensor element, ε_j is the responding symmetric strain tensor element, and C_{ij} (*i* and *j* are the Voigt deformation components, equal to 1, 2, 3, 4, 5, and 6) is the elastic constant tensor. The zero energy minimization method was applied before the deformation. The compliance matrix (*S*) was later derived from the inverse of the elastic constant matrix ($S = C^{-1}$). The compliance and elastic constant matrices were used in Voigt, Reuss, and Hill's methods to obtain bulk modulus (*B*) and shear modulus (*G*). The Voigt method is based on the aggregation of uniform strain, as shown in eqs 5 and 6.

$$B_{\text{Voigt}} = \frac{1}{9}(C_{11} + C_{22} + C_{33} + 2(C_{12} + C_{13} + C_{23})) \quad (5)$$

$$G_{\text{Voigt}} = \frac{1}{15}(C_{11} + C_{22} + C_{33} - (C_{12} + C_{13} + C_{14}) + 3(C_{44} + C_{55} + C_{66})) \quad (6)$$

The Reuss method is based on the aggregation of the uniform stress, as shown in eqs 7 and 8.

$$B_{\text{Reuss}} = (S_{11} + S_{22} + S_{33} + 2(S_{31} + S_{21} + S_{32}))^{-1} \quad (7)$$

$$G_{\text{Reuss}} = 15/[4(S_{11} + S_{22} + S_{33} - S_{12} - S_{23} - S_{13}) + 3(S_{44} + S_{55} + S_{66})] \quad (8)$$

Young's modulus is calculated by eq 9.

$$E = \left(\frac{1}{3G} + \frac{1}{9K} \right)^{-1} \quad (9)$$

For statistical accuracy, the bulk modulus, the shear modulus, and Young's modulus reported in this study were calculated using Hill's method, which is the geometric mean of the moduli calculated from Voigt and Reuss methods.⁴² Before calculating the final result of mechanical properties, parameters such as numbers of iterations of force/energy minimization and evaluation of each glass compositions were toggled until the convergence of the elastic constant tensor (C_{11} , C_{22} , or C_{33}) was reached to ensure the validity of the calculation. Also, the trajectories from six directions were examined to ensure

that no fracture occurs in the simulation box within the deformation process.

2.4. QSPR Analysis. We use a structural descriptor (F_{net}) that was first developed by Lusvardi et al.^{30,32} to summarize the structural characteristics and chemical bonding contributions. The descriptor (with some modification) has been used in our earlier work on the zirconia effect on borosilicate glasses to describe the density and dissolution rate changes with the composition and thermomechanical properties such as thermal expansion coefficients, hardness, and Young's moduli of calcia aluminosilicate glasses, all in combination with structural characteristics from MD simulations. The mathematical expression is shown by eq 10.

$$F_{\text{net}} = \frac{1}{N_{\text{atoms}}} \sum_x^{N_{\text{cations}}} n_x \text{CN}_{x\text{O}} \text{SBS}_{x\text{O}} m_x \quad (10)$$

where N_{atoms} is the total number of atoms, N_{cations} is the total number of cations, n_x is the number of cation x ($x = \text{Na}, \text{Ca}, \text{Al}, \text{B}^{[4]}, \text{B}^{[3]}, \text{Si}, \text{and Zr}$), $\text{CN}_{x\text{O}}$ is the average coordination number (CN) of "cation x –oxygen pair", $\text{SBS}_{x\text{O}}$ is the single bond strength obtained from the study by Sun,⁴³ and lastly, m_x is the multiplicative factor for each cation–oxygen pair. It is worth noting that the earlier QSPR analysis used the bond strength of a diatomic molecule or cation, but our studies showed that the single bond strength proposed by Sun and used for classification of glass formers and modifiers can provide a consistent trend in wide ranges of glass compositions and various properties. Hence, the single bond strength was used in this work for QSPR analysis of the original and ion-exchanged glasses. Table 4 below summarizes the $\text{SBS}_{x\text{O}}$ and m_x values used in this study.

Table 4. Single Bond Strength and Multiplicative Factor of Cation–Oxygen Pairs

cation–oxygen pair	single bond strength (kcal/mol)	m_x (multiplicative factor)
Na–O	20	1
K–O	13	1
Ca–O	32	2
$\text{B}^{[3]}$ –O	119	3
Al–O	79	4
$\text{B}^{[4]}$ –O	89	4
Si–O	106	4
Zr–O	81	6

3. RESULTS

3.1. Influence of Composition and Synthesis Methods on the Mechanical Properties. Figures 3 and 4 show Young's, bulk, and shear modulus values calculated for the (IOX)ISG-an and (IOX)ISG-mq glass, respectively. From Figures 3 and 4, three observations regarding the effects of glass former substitution, IOX concentration, and synthesis method can be made: (1) the mechanical properties decrease with the increasing amount of $\text{B}_2\text{O}_3 + \text{Na}_2\text{O}/\text{SiO}_2$, using both annealing and melting-and-quenching methods; (2) the mechanical moduli do not deviate much from that of 0% K/Na glass compositions, suggesting that IOX has a limited effect on the mechanical properties of ISG series glass; and (3) both Figures 3 and 4 show a flat curve with the increasing IOX concentration, suggesting that $\text{B}_2\text{O}_3 + \text{Na}_2\text{O}/\text{SiO}_2$ concentration has a higher effect on the mechanical modulus than Na/K substitution.

Figures 5 and 6 show the mechanical moduli calculated for NQ-an and NQ-mq series of glasses, respectively. Three observations can be made: (1) The mechanical properties decrease as the $\text{Al}_2\text{O}_3 + \text{Na}_2\text{O}/\text{SiO}_2$ substitution increases for both annealing and melting-and-quenching methods. (2) The three mechanical moduli show a steady increase with the increasing K/Na concentration; moreover, this feature is not affected by the $\text{Al}_2\text{O}_3 + \text{Na}_2\text{O}/\text{SiO}_2$ substitution. (3) Comparing between Figures 5 and 6, the increasing trend of mechanical properties with the IOX concentration increase is diminished by the melting-and-quenching method. The observation indicates that the influence of IOX could be hindered by the relaxation from high-temperature melting.

As a short summary, the mechanical moduli of NQ series glasses are positively correlated to Na/K concentration, and the high temperature from melting-and-quenching method would lower the value via structural relaxation. On the other hand, $\text{B}_2\text{O}_3 + \text{Na}_2\text{O}/\text{SiO}_2$ has more influence on the mechanical moduli of ISG series glasses than K/Na concentration. However, it is important to note that mechanical moduli alone cannot be considered as an indication of compressive stress generated by the IOX process.

3.2. QSPR Analysis of the Mechanical Properties. The correlation between the mechanical properties and the structural descriptor F_{net} of (IOX)ISG-an and (IOX)ISG-mq glasses are established and shown in Figures 7 and 8, respectively. Both Figures 7 and 8 show a good linear correlation between mechanical moduli and F_{net} . Figure 7 shows that the data points can be roughly grouped into three regions according to the values of F_{net} which are ~ 480 , ~ 490 ,

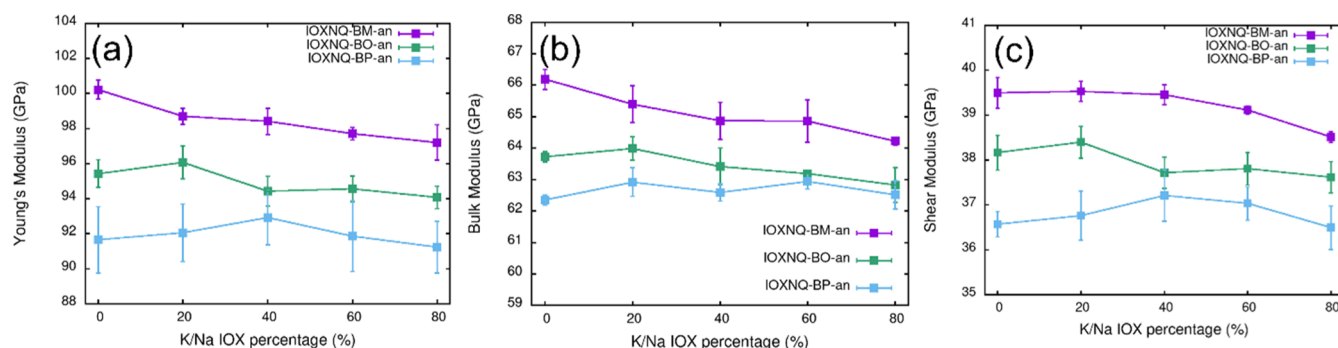


Figure 3. (a) Young's, (b) bulk, and (c) shear moduli of the (IOX)ISG-an (annealed glass) series glass with increasing IOX concentration.

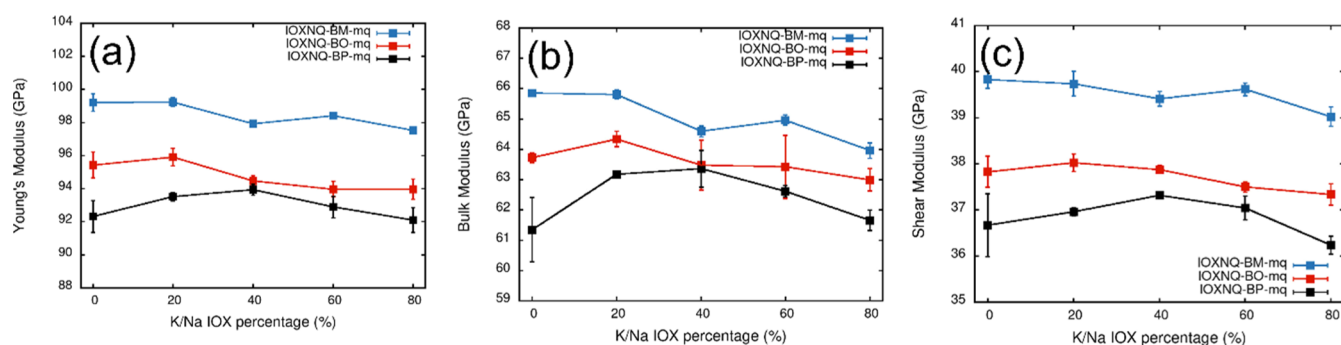


Figure 4. (a) Young's, (b) bulk, and (c) shear moduli of IOXISG-mq (melted and quenched glass) series glass with increasing IOX concentration.

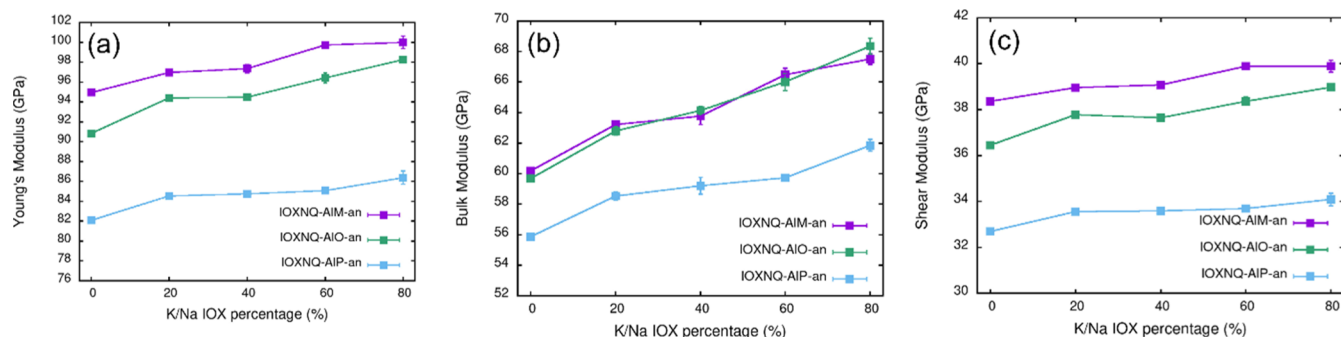


Figure 5. (a) Young's, (b) bulk, and (c) shear moduli of the IOXNQ-an (annealed) series glass with increasing IOX concentration.

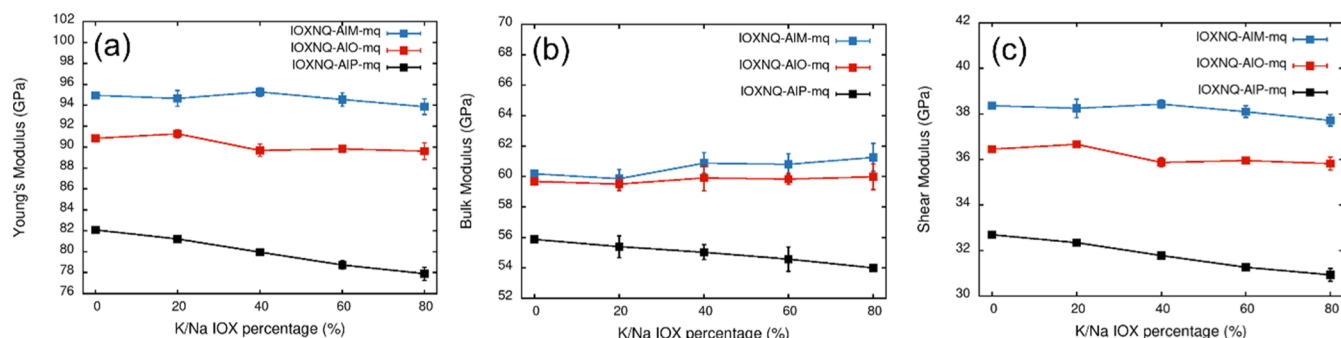


Figure 6. (a) Young's, (b) bulk, and (c) shear moduli of IOXNQ-mq (melted and quenched) series glass with increasing IOX concentration.

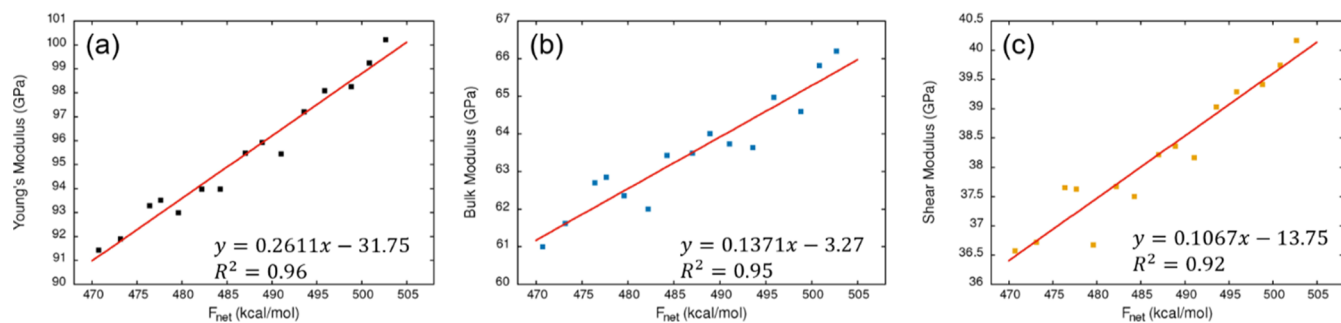


Figure 7. Correlation between the structural descriptor (F_{net}) and (a) Young's, (b) bulk, and (c) shear moduli of (IOX)ISG-an series glass.

and ~ 500 kcal/mol. As listed in Table 5, these regions correspond to ISG-BP, ISG-BO, and ISG-BM, respectively. The result shows that the mechanical properties of the ISG series glasses have a strong linear correlation with the structural descriptor (F_{net}). Additionally, for glasses generated via the melting-and-quenching method, as shown in Figure 8, the

correlation coefficient between mechanical properties and F_{net} is improved in (IOX)ISG-mq glasses.

Figures 9 and 10 show the correlation between the F_{net} and the mechanical properties of (IOX)NQ-an and (IOX)NQ-mq glasses, respectively. The result of QSPR fitting of (IOX)NQ series glass and (IOX)ISG series glass show a much poorer linear correlation, especially in comparison to those in Figures

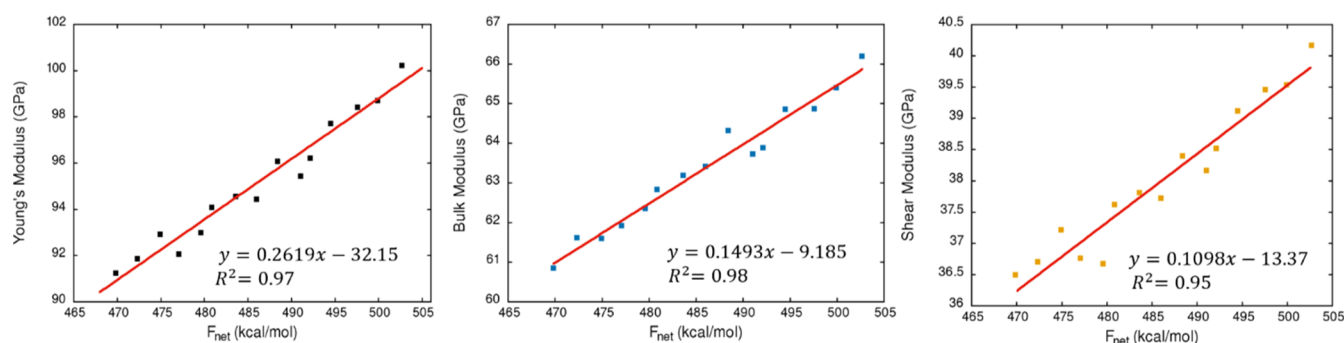


Figure 8. Correlation between the structural descriptor (F_{net}) and (a) Young's, (b) bulk, and (c) shear moduli of (IOX)ISG-mq series glass.

Table 5. Theoretical Structural Descriptor (F_{net}) of Non-IOX Glasses

glass composition	F_{net} (kcal/mol)
ISG-BM	499.91 $\pm$ 0.15
ISG-BO	488.41 $\pm$ 0.57
ISG-BP	477.08 $\pm$ 0.48
NQ-AIM	499.97 $\pm$ 0.08
NQ-AIO	472.91 $\pm$ 0.20
NQ-AIP	439.93 $\pm$ 0.16

7 and 8 where good linear correlations were found for both the (IOX)ISG-an and -mq series of glasses. As shown in Figure 9, the data points can be distinctively grouped into three regions according to the values of F_{net} , which are ~ 440 , ~ 470 , and ~ 500 kcal/mol, as listed in Table 5. The detailed structural contribution of F_{net} can be found in Tables S1 and S2. From low to high F_{net} , these three regions correspond to NQ-AIP, NQ-AIO, and NQ-AIM, respectively. Within each region, it can be seen that increasing the K^+ -to- Na^+ IOX ratio results in significant increase in elastic moduli but with only a small change in F_{net} . Comparing Figures 9 and 10, the correlation between F_{net} and mechanical properties of the (IOX)NQ-mq glasses is more linear than that for the (IOX)NQ-an series of glasses. These results suggest that in the ion-exchanged NQ series glasses after annealing, there are relatively small structural changes, as shown in similar F_{net} and vertical grouping of different levels of IOX for each glass composition (Figure 9). This is in contrast to ISG series glasses (Figure 7) where the annealed IOX glasses show relatively good linear correlations between mechanical properties and F_{net} . Together with later analysis of LNDC, this abnormal behavior of QSPR correlation of annealed IOX glasses can be an indication of good candidate for IOX strengthening.

3.3. LNDC and Compressive Stress. The final molar volumes of ISG and NQ glasses generated by annealing and melting-and-quenching as the function of IOX concentration are shown in Figures 11 and 12, respectively. Figure 11 shows that the molar volume of all ISG series glass increases as the IOX concentration increases. Figure 11a–c shows the difference between the molar volume of glass acquired from the two methods shrink as $\text{B}_2\text{O}_3 + \text{Na}_2\text{O}/\text{SiO}_2$ substitution increases. Furthermore, the increase of molar volume of melted-and-quenched glass is at a rate approximately 2 times that of the annealed glass, which implies 2 times the strain and LNDC. Figure 11 also shows some deviation from the linear function. Since the values are small, we consider the deviation to be resulted from the lack of parallel tests and the time of melting or equilibrium in MD simulation, which have a high influence when investigating multicomponent glasses.

Figure 12 shows that the increase of $\text{Al}_2\text{O}_3 + \text{Na}_2\text{O}/\text{SiO}_2$ substitution has little effect on the molar volume of NQ series glass. The increase of molar volume of NQ-mq glasses with the increasing IOX concentration is also at a rate approximately 2 times that of NQ-an glasses, which is similar to the behavior in ISG series glass. Comparing Figures 12 and 11, the data deviation is smaller than that for ISG series glass, which gives the overall linear fitting a R^2 value larger than 0.9. One possible explanation is that the composition of NQ series glass is simpler than that of ISG series glass, which allows a more prominent effect of K/Na substitution in the NQ series than that in the ISG series.

The LNDCs of ISG and NQ series glasses were calculated by using the molar volume difference between annealed and melted-and-quenched glasses as a function of IOX concentration ($\partial V/\partial C$ in eq 2), and the results are shown in Figure 13. Importantly, the LNDC induced in the NQ series is 3–4 times higher than those of the ISG series, suggesting that the

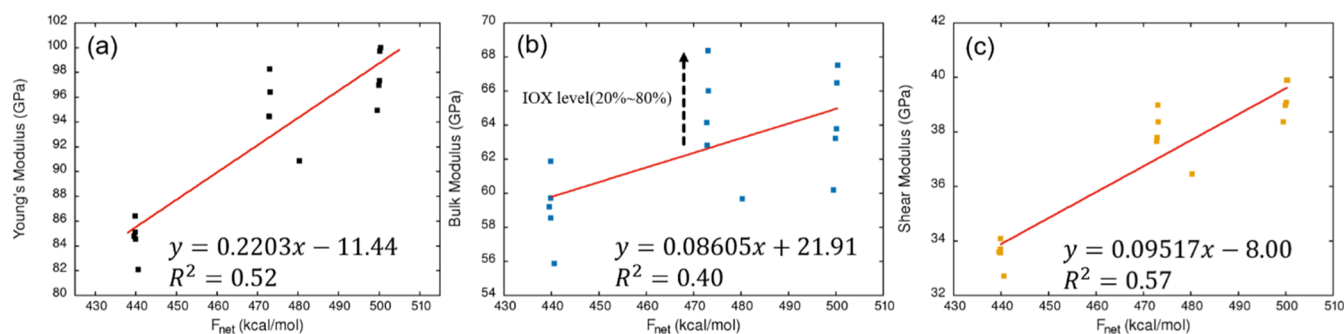


Figure 9. Correlation between the structural descriptor (F_{net}) and (a) Young's, (b) bulk, and (c) shear moduli of (IOX)NQ-an series glass.

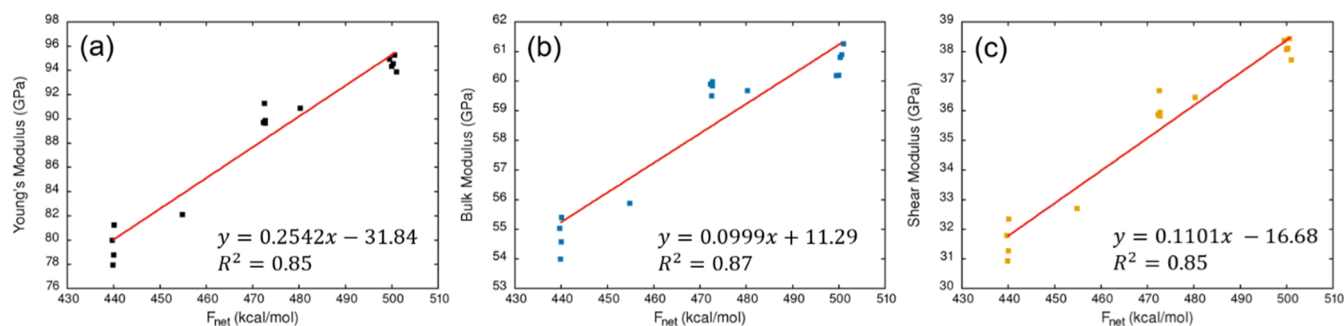


Figure 10. Correlation between the structural descriptor (F_{net}) and (a) Young's, (b) bulk, and (c) shear moduli of (IOX)NQ-mq series glass.

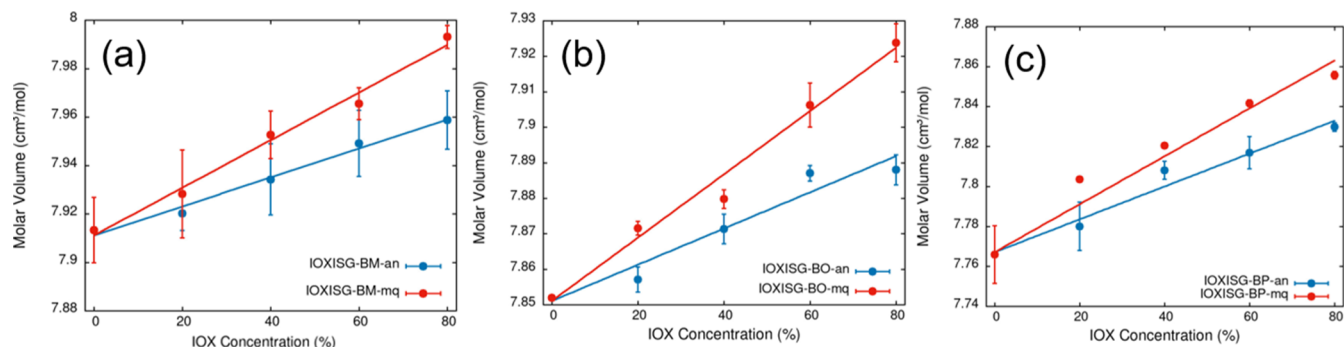


Figure 11. Influence of IOX on molar volume for the annealing and melting-and-quenching methods in (a) ISG-BM, (b) ISG-BO, and (c) ISG-BP glasses.

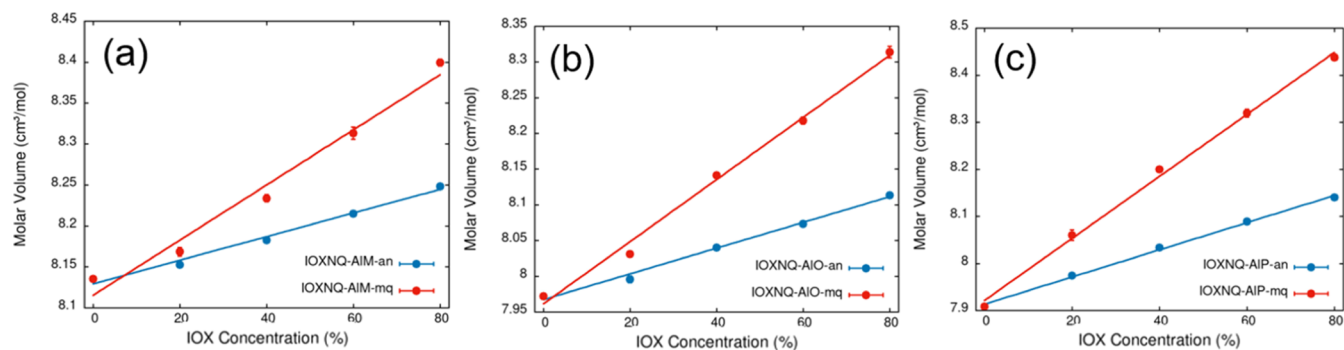


Figure 12. Influence of IOX on molar volume for the annealing and melting-and-quenching methods in (a) NQ-AIM, (b) NQ-AIO, and (c) NQ-AIP glass.

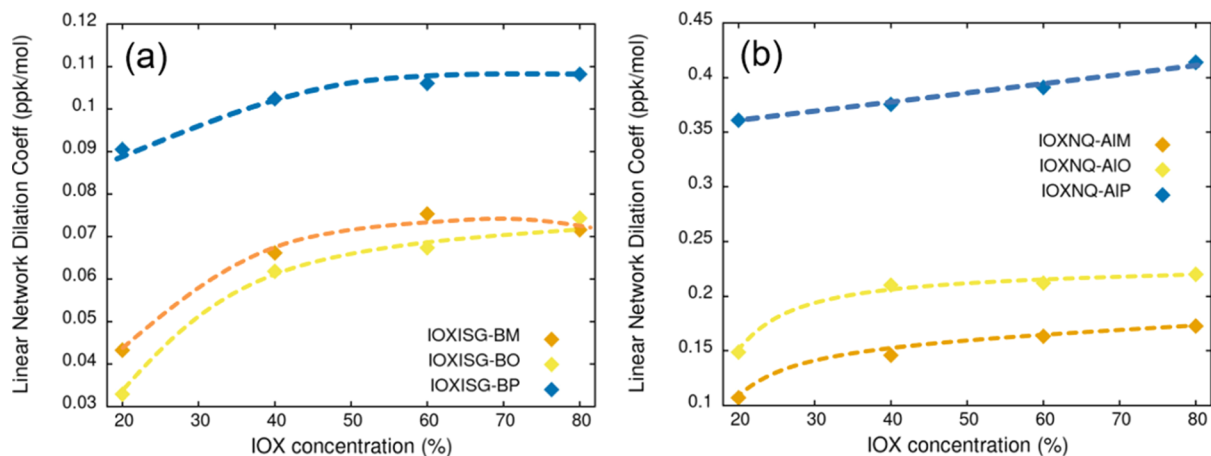


Figure 13. LNDC of (a) ISG series glass and (b) NQ series glass as a function of increasing K/Na substitution.

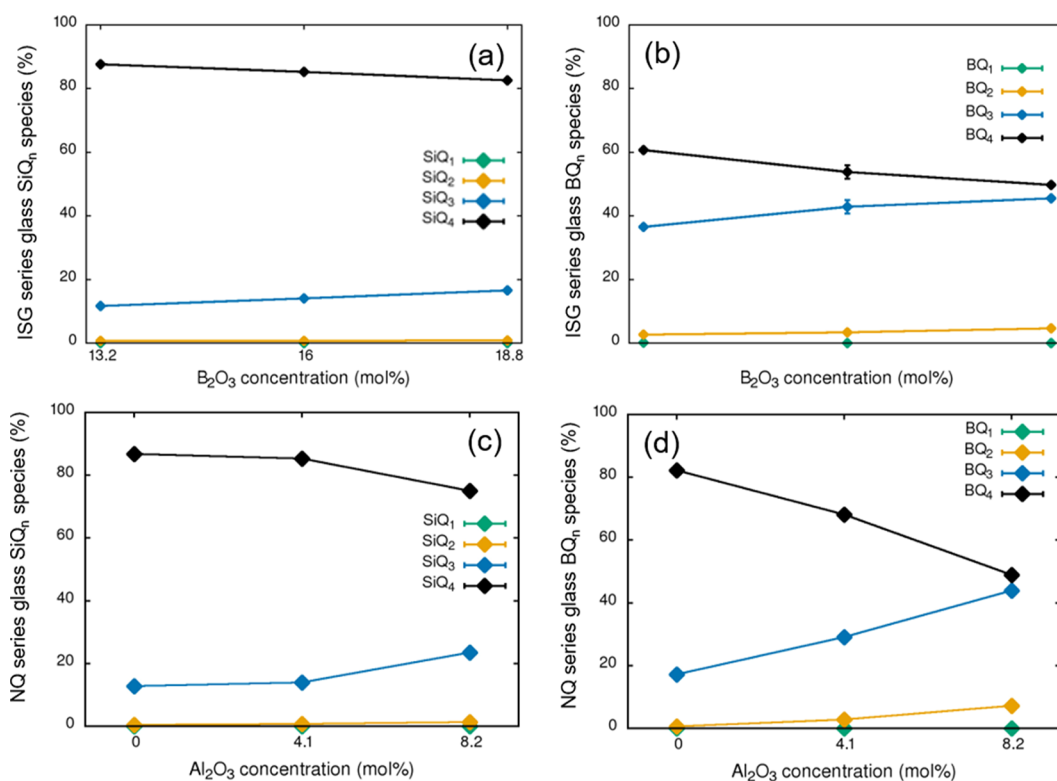


Figure 14. Q_n species of (a) Si and (b) B as a function of $B_2O_3 + Na_2O/SiO_2$ in ISG series glass. Q_n species of (a) Si and (b) as a function of $Al_2O_3 + Na_2O/SiO_2$ in NQ series glass.

NQ series glasses are more suitable for IOX strengthening than the ISG series. Additionally, the composition with the highest amount of glass formers (IOXISG-BP and IOXNQ-AIP) have the largest LNDC. Comparing Figure 13a,b, the LNDCs of IOXISG and IOXNQ series glass share two similar behaviors: (1) the LNDC increases logarithmically, instead of linearly, with IOX concentration, which suggests that the strain induced by each mole of K/Na substitution is lower than the strain resulted from the previous substitution, resulting to a saturation point of strain. (2) The baseline of the LNDC is mainly determined by the compositions, meaning that the amount of strain induced by IOX is highly affected by the glass former substitution. In Figure 13a, the LNDC curves of IOXISG-BO and IOXISG-BM are similar and share the same saturation point ($B \sim 0.073$ ppk/mol of K/Na) with each other. This behavior is similar to Figure 13b, where the LNDCs of IOXNQ-AIM-an and IOX-AIO-an are close with each other. Furthermore, Figure 13a,b shows that the LNDC values of IOXISG-BP-an and IOXNQ-AIP-an are higher than those of the other two glasses in the same series. The result suggests that more strain can be induced from IOX for the higher amount of $Al_2O_3 + Na_2O/SiO_2$ in NQ series glass and higher amount of $B_2O_3 + Na_2O/SiO_2$ in ISG series glass.

4. DISCUSSION

The results from MD simulations show that mechanical properties and LNDC of glasses are affected by both the glass former substitution and IOX concentration. Some behaviors corresponding to the compositional change can be found (1) the mechanical moduli shown in section 3.1 show three different baselines for increasing glass former substitution. Removing SiO_2 lowers the mechanical properties even with the increase of glass formers per substitution [i.e., $(B_2O_3 + Na_2O)/$

SiO_2 for ISG series glass and $(Al_2O_3 + Na_2O)/SiO_2$ for NQ series glass]. (2) Despite both ISG and NQ being borosilicate-based compositions, the enhancement of mechanical properties by IOX is more evident in the NQ series glass than that in ISG series glass, which can be resulted from the compositional difference. (3) The F_{net} of (IOX)NQ series glass can be divided into three different regions, while no obvious data separation can be found in (IOX)ISG series glass, suggesting different abilities of accommodating substituted K^+ .

Since the structural features have a prominent effect on the mechanical properties, the change of Q_n species of major glass formers with glass former substitution were calculated. Here Q_n has refers to the glass former cation polyhedron with n bridging oxygen atoms. Originally, Q_n is specifically used for glass former cations with four-fold oxygen coordination (Q). For the complex glasses studied in this work, we have glass formers that have a CN other than four, but for simplicity, we denote these as SiQ_n or BQ_n species. Particularly, we consider SiO_2 , Al_2O_3 , B_2O_3 , and ZrO_2 as glass formers so an oxygen that is linked with any of two of these glass former cations is considered to be a bridging oxygen. Figure 14a,b shows that SiQ_4 transforms into SiQ_3 ($SiQ_4 \rightarrow SiQ_3$) and BQ_4 transforms into BQ_3 ($BQ_4 \rightarrow BQ_3$), while AlQ_n and ZrQ_n do not change much as the $B_2O_3 + Na_2O/SiO_2$ increases in ISG series. For NQ series glass, Figure 14c,d shows the same behavior of $SiQ_4 \rightarrow SiQ_3$ and $BQ_4 \rightarrow BQ_3$ with increasing $Al_2O_3 + Na_2O/SiO_2$. This decrease in the Q_n species of major glass formers in ISG and NQ series glass implies the decrease in the network connectivity and mechanical properties, which explains why the baseline of mechanical moduli shown in Figure 3 and 5 shifts lower as the compositional substitution increases.

To explain the different behaviors of mechanical moduli of ISG and NQ series glass with the increasing IOX

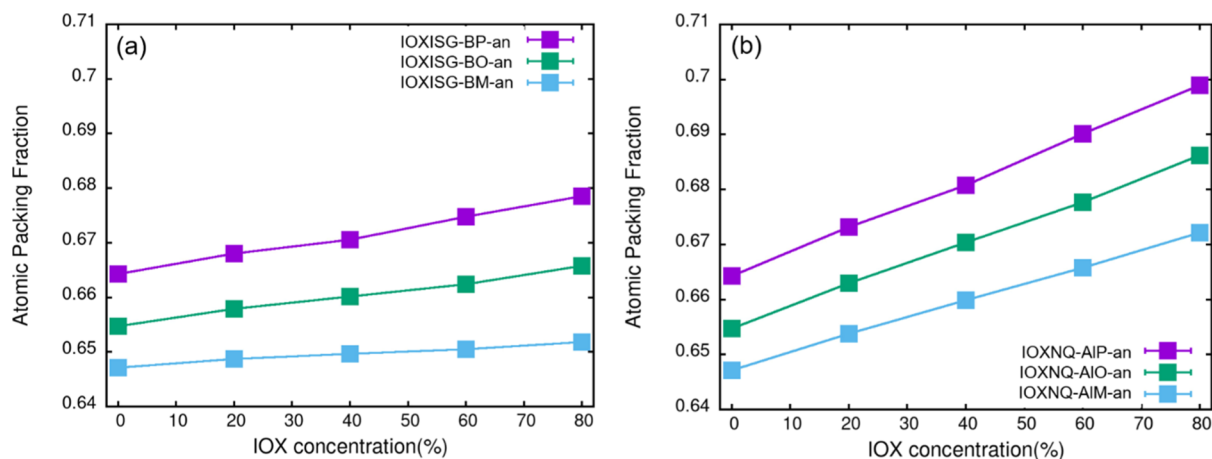


Figure 15. APF of (a) (IOX)ISG series glass and (b) (IOX)NQ series glass.

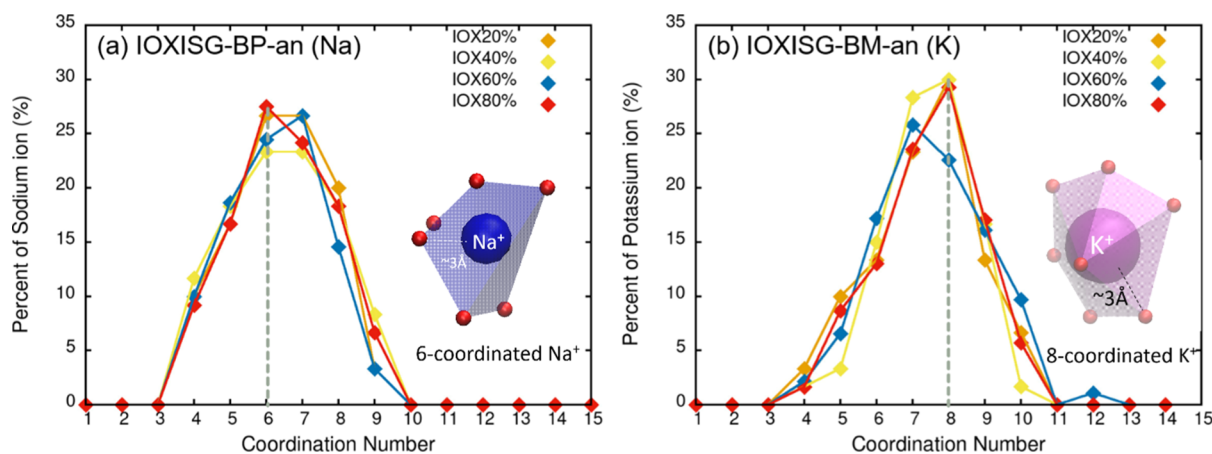


Figure 16. Distribution of (a) Na and (b) K ion CNs and their local environment (within 3 Å) of IOXISG-BP-an glass.

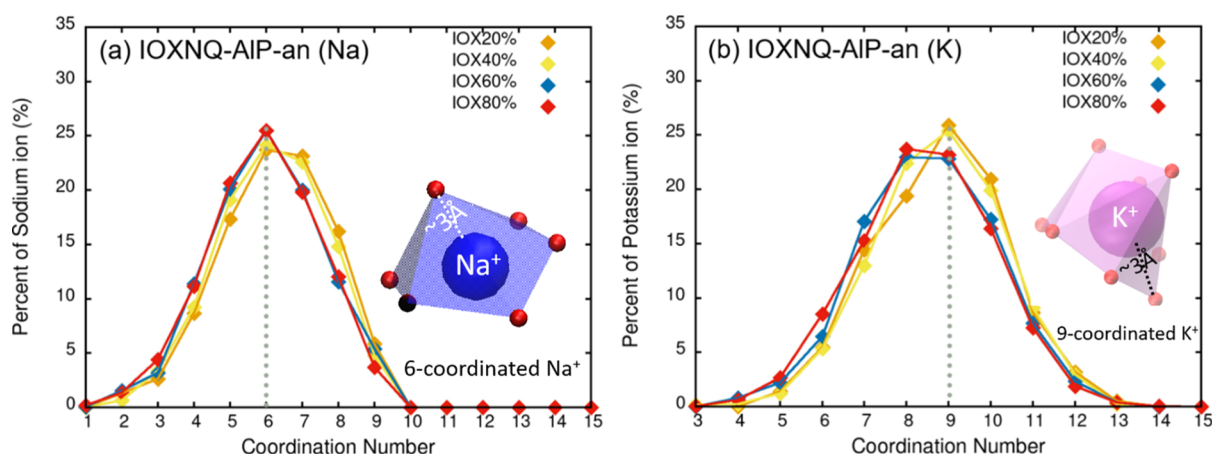


Figure 17. Distribution of (a) Na and (b) K ion CNs and their local environment (within 3 Å) of IOXNQ-AIM-an glass.

concentration, we looked into the change in the atomic packing fraction (APF). The APF is factor that represents how “packed” the atom is in the cell, which can be considered as the amount of interatomic action required to induce strain. The mathematical equation of the APF is given by eq 11

$$\text{APF} = \frac{\sum_{x=1}^{N_{\text{ion}}} \frac{4}{3} \pi R_x^3 n_x}{V_{\text{cell}}} \quad (11)$$

where V_{cell} is the cell volume, R_x is Shannon's radii of ion x ($x = \text{Na}, \text{Ca}, \text{Al}, \text{B}^{[4]}, \text{B}^{[3]}, \text{Si}, \text{and Zr}$), and n_x is the number of x ions in the glass. Comparing Figure 15a,b, the APFs of non-IOX ISG and NQ series glass are similar; however, the APF of NQ series glass increases slightly more significantly than the APF of ISG series glass with the increasing IOX concentration. The result explains the more prominent increase of mechanical moduli by IOX in NQ series glass than that in ISG series glass.

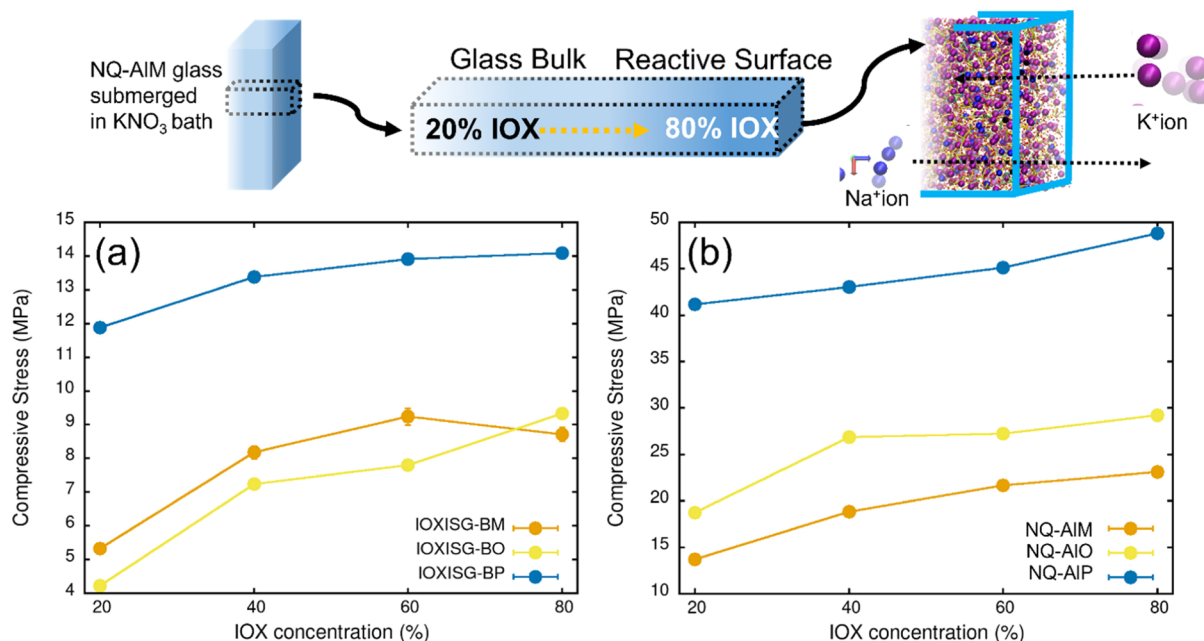


Figure 18. Schematics for the IOX process of a glass and the compressive profiles of (a) IOXISG and (b) IOXNQ series glass.

The difference in the mechanical moduli of ISG and NQ series glass can also be explained by the difference in the CN distribution of modifier cations K^+ and Na^+ . As shown in Figures 16 and 17, the peak of K CN distribution in ISG-BP glass is lower than that in NQ-AIP glass, showing that the local environment of K ions is more compact in NQ-AIP than that in ISG-BP glass, hence benefitting the enhancement of mechanical properties in NQ series glass. On the other hand, CN distribution of K and Na does not vary much with the IOX concentration in ISG series glass, which also corresponds to the limited change in the APF with IOX (Figure 15a vs 15b). The different behaviors of alkali ion CN distribution as a function of IOX concentration suggests that the structures of the multicomponent ISG series glasses are more complex and have more factors involved than the NQ series. At this point, we can deduce that the glass structure of ISG series glasses are more structurally flexible than that of the NQ series glasses; hence, the ISG series glasses are capable of accommodating the substituted K^+ ions and allowing them to move into positions of existing Na^+ positions without large local structural relaxations. This can be seen from parameters used for F_{net} calculations listed in Tables S1 and S2. This can also explain the different trend of F_{net} in IOXISG and IOXNQ glasses (Figure 7 and 8), which are mainly contributed by the change of K^+ and Na^+ contributions instead of the local structural changes of the glass former cations (as shown in Tables S1 and S2).

Based on the stress generated after IOX from simulations and neglecting stress relaxation, we obtained the stress profile of IOX glasses. The compressive stresses generated for each glass as a function of the level of IOX were calculated using eq 1, and the results are shown in Figure 18. The compressive stress gradually increases with the increasing K/Na concentration and shows a plateau when the IOX concentration reaches 40–60%. Moreover, similar to elastic modulus, the value of compressive stress is influenced by the glass former substitution, that is, the overall SiO_2 concentration. Compared to the compressive stress after IOX for commercially available

Gorilla glass (usually in the 500–800 MPa range), the compressive stress values of ISG and NQ series glass are significantly lower. Furthermore, although Young's moduli of (IOX)ISG and (IOX)NQ series glasses are in the same range of 80–100 GPa (Figures 3 and 5), the compressive stress of IOXNQ series glass is significantly larger than that of the IOXISG series glass. Young's moduli calculated using MD simulations agree well with experimental values; hence, the low compressive stress after IOX is mainly due to the low LNDC that is about 10 times lower than the experimental values.^{16,44} This also indicates the importance of the LNDC for the compressive stress generated by chemical strengthening. Importantly, the MD simulation results from this work show that the LNDC strongly depends on glass composition and it cannot be considered as simply a second-order effect in the multicomponent borosilicate glass. It is important to note that in MD simulation, the small size of the simulation cell can result in a certain degree of compositional inhomogeneity, which can affect the C_{avg} in eq 1. Furthermore, it would be valuable to calculate the fracture toughness from IOX-processed glass using MD simulations as that can provide further insights into the strengthening of the glass materials.

5. CONCLUSIONS

This study explored the glass composition effect on IOX strengthening of borosilicate glasses by using MD-based atomistic simulations and QSPR analyses. We adopted two different glass former substitutions: $B_2O_3 + Na_2O/SiO_2$ substitution was performed on one of the series of glass and $Al_2O_3 + Na_2O/SiO_2$ substitution on the other series. For each glass composition, IOX was studied with 20, 40, 60, and 80% of Na^+ being randomly substituted by K^+ to study the effect of IOX. Structural features of these glasses, before and after IOX, and various mechanical properties were calculated. Two different methods, annealing (-an) and melting-and-quenching (-mq), were used to generate the IOX glasses. The relationship between the mechanical properties and the theoretical structural descriptor (F_{net}) was established through QSPR

analysis. The volume strain induced by IOX is derived by the molar volume difference between the annealed and melted-and-quenched glasses, based on which the LNDC, a critical parameter for IOX, was calculated.

The LNDC calculated from MD simulations shows that LNDC values are 3–4 times higher in the NQ series of glasses than that in the ISG series of glasses. Also, the glasses with the highest amount of mixed glass formers show the highest LNDC. As the surface compressive stress profile is largely determined by the magnitude of the LNDC, NQ series of glasses show 3–4 times higher surface compressive stress than the ISG series. As a result, the NQ series of glass is much more efficient and suitable for IOX strengthening than the ISG series, which is a remarkable result from our MD simulations and IOX analysis. The results also suggest that the capability of enhancing mechanical properties by IOX could be diminished by the increase of compositional complexity and decrease of network connectivity in the multicomponent borosilicate glasses.

To understand the different influences of IOX on ISG and NQ series glass, detailed structural analysis using MD-generated models was performed and the APFs were calculated. The APF of the NQ series glass increases more significantly than that of ISG series glass as a function of K/Na substitution. Analysis of the local environment of modifier cations shows that K^+ can be quickly accommodated in the IOX samples of ISG glass series, while K^+ in IOX samples of NQ series of glass show a similar local environment to Na^+ . This can be explained by the fact that the higher composition complexity in the ISG glass series with four oxides (SiO_2 , B_2O_3 , Al_2O_3 , and ZrO_2) can play the role of glass formers. On the other hand, the glass network in the NQ series glass series is less accommodating, which forces K^+ ions to reside in similar environments as Na^+ , which leads to an increase of the APF and high compressive stress. This is supported by the observation of lower strain induced and less mechanical property change introduced by IOX in the ISG series glasses.

This study thus shows that combining QSPR analysis and the LNDC calculated from MD simulations, the effectiveness of IOX to a glass composition can be predicted and the strengthening mechanism elucidated for complex glass compositions. They are thus a very effective way of investigating IOX strengthening of glass materials.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcb.1c10928>.

Multiplicative factor of glass former cations for F_{net} calculations and CN distributions of cations for the three series of glass at different levels of K^+/Na^+ IOX from MD simulations (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Jincheng Du – Department of Material Science and Engineering, University of North Texas, Denton, Texas 76203, United States; orcid.org/0000-0003-4805-7498; Email: du@unt.edu

Author

Po-Hsuen Kuo – Department of Material Science and Engineering, University of North Texas, Denton, Texas 76203, United States

Complete contact information is available at:

<https://pubs.acs.org/doi/10.1021/acs.jpcb.1c10928>

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

The authors gratefully acknowledge the support of the National Science Foundation (NSF) (project #1662288). J.D. also would like to thank the Gordon S. Fulcher Sabbatical Program and helpful discussions with Dr. Doug Allan of Corning Inc. Computational resources were provided by the UNT Talon3 HPC cluster.

■ REFERENCES

- (1) Kistler, S. S. Stress in Glass Produced by Nonuniform Exchange of Monovalent Ions. *J. Am. Ceram. Soc.* **1962**, *45*, 59–68.
- (2) Gy, R. Ion Exchange for Glass Strengthening. *Mater. Sci. Eng., B* **2008**, *149*, 159–165.
- (3) Kawamura, H.; Suzuki, R.; Matsusaka, S.; Hidai, H.; Chiba, A.; Morita, N. Formation of an Embedded Electrical Circuit in Glass Substrate by Solid-State Ion Exchange with Application of a Forward/Reverse Voltage. *Precis. Eng.* **2019**, *55*, 240–247.
- (4) Varshneya, A. K. Chemical Strengthening of Glass: Lessons Learned and Yet To Be Learned. *Int. J. Appl. Glass Sci.* **2010**, *1*, 131–142.
- (5) Nordberg, M. E.; Mochel, E. L.; Garfinkel, H. M.; Olcott, J. S. Strengthening by Ion Exchange. *J. Am. Ceram. Soc.* **1964**, *47*, 216–219.
- (6) Varshneya, A. K. *Fundamentals of Inorganic Glasses*; Elsevier, 2013.
- (7) Cooper, A. R.; Krohn, D. A. Strengthening of Glass Fibers: II, Ion Exchange. *J. Am. Ceram. Soc.* **1969**, *52*, 665–669.
- (8) Sane, A. Y.; Cooper, A. R. Stress Buildup and Relaxation During Ion Exchange Strengthening of Glass. *J. Am. Ceram. Soc.* **1987**, *70*, 86–89.
- (9) Varshneya, A. K.; Milberg, M. E. Ion exchange in sodium borosilicate glasses. *J. Am. Ceram. Soc.* **1974**, *57*, 165.
- (10) Soules, T. F.; Varshneya, A. K. Molecular Dynamic Calculations of A Sodium Borosilicate Glass Structure. *J. Am. Ceram. Soc.* **1981**, *64*, 145–150.
- (11) Varshneya, A. K.; Olson, G. A.; Kreski, P. K.; Gupta, P. K. Buildup and Relaxation of Stress in Chemically Strengthened Glass. *J. Non-Cryst. Solids* **2015**, *427*, 91–97.
- (12) Macrelli, G.; Varshneya, A. K.; Mauro, J. C. Simulation of Glass Network Evolution during Chemical Strengthening: Resolution of the Subsurface Compression Maximum Anomaly. *J. Non-Cryst. Solids* **2019**, *522*, 119457.
- (13) Richmond, O. Theory of Residual Stress Due to Chemical Concentration Gradients. *Trans. ASM Q* **1964**, *57*, 294.
- (14) Fu, A. I.; Mauro, J. C. Mutual Diffusivity, Network Dilation, and Salt Bath Poisoning Effects in Ion-Exchanged Glass. *J. Non-Cryst. Solids* **2013**, *363*, 199–204.
- (15) Varshneya, A. K. The Physics of Chemical Strengthening of Glass: Room for a New View. In *J. Non-Cryst. Solids* **2010**, *356*, 2289–2294.
- (16) Vargheese, K. D.; Tandia, A.; Mauro, J. C. Molecular Dynamics Simulations of Ion-Exchanged Glass. *J. Non-Cryst. Solids* **2014**, *403*, 107–112.
- (17) Bechgaard, T. K.; Goel, A.; Youngman, R. E.; Mauro, J. C.; Rzoska, S. J.; Bockowski, M.; Jensen, L. R.; Smedskjaer, M. M. Structure and Mechanical Properties of Compressed Sodium

Aluminosilicate Glasses: Role of Non-Bridging Oxygens. *J. Non-Cryst. Solids* **2016**, *441*, 49–57.

(18) Tandia, A.; Vargheese, K. D.; Mauro, J. C. Elasticity of Ion Stuffing in Chemically Strengthened Glass. *J. Non-Cryst. Solids* **2012**, *358*, 1569–1574.

(19) Tandia, A.; Vargheese, K. D.; Mauro, J. C.; Varshneya, A. K. Atomistic Understanding of the Network Dilation Anomaly in Ion-Exchanged Glass. *J. Non-Cryst. Solids* **2012**, *358*, 316–320.

(20) Mauro, J. C.; Gupta, P. K.; Loucks, R. J. Composition Dependence of Glass Transition Temperature and Fragility. II. A Topological Model of Alkali Borate Liquids. *J. Chem. Phys.* **2009**, *130*, 234503.

(21) Katritzky, A. R.; Lobanov, V. S.; Karelson, M. QSPR: the correlation and quantitative prediction of chemical and physical properties from structure. *Chem. Soc. Rev.* **1995**, *24*, 279–287.

(22) Kholodovych, V.; Smith, J. R.; Knight, D.; Abramson, S.; Kohn, J.; Welsh, W. J. Accurate Predictions of Cellular Response Using QSPR: A Feasibility Test of Rational Design of Polymeric Biomaterials. *Polymer* **2004**, *45*, 7367–7379.

(23) Yu, X.; Wang, X.; Li, X.; Gao, J.; Wang, H. Prediction of Glass Transition Temperatures for Polystyrenes by a Four-Descriptors QSPR Model. *Macromol. Theory Simul.* **2006**, *15*, 94–99.

(24) Cypcar, C. C.; Camelio, P.; Lazzeri, V.; Mathias, L. J.; Waegell, B. Prediction of the Glass Transition Temperature of Multicyclic and Bulky Substituted Acrylate and Methacrylate Polymers Using the Energy, Volume, Mass (EVM) QSPR Model. *Macromolecules* **1996**, *29*, 8954–8959.

(25) Mattioni, B. E.; Jurs, P. C. Prediction of Glass Transition Temperatures from Monomer and Repeat Unit Structure Using Computational Neural Networks. *J. Chem. Inf. Comput. Sci.* **2002**, *42*, 232–240.

(26) Pilania, G.; Iverson, C. N.; Lookman, T.; Marrone, B. L. Machine-Learning-Based Predictive Modeling of Glass Transition Temperatures: A Case of Polyhydroxyalkanoate Homopolymers and Copolymers. *J. Chem. Inf. Model.* **2019**, *59*, 5013–5025.

(27) Santos, C. B. R.; Lobato, C. C.; Braga, F. S.; Morais, S. S. S.; Santos, C. F.; Fernandes, C. P.; Brasil, D. S. B.; Hage-Melim, L. I. S.; Macêdo, W. J. C.; Carvalho, J. C. T. Application of Hartree-Fock Method for Modeling of Bioactive Molecules Using SAR and QSPR. *Comput. Mol. Biosci.* **2014**, *4*, 1–24.

(28) Cravero, F.; Martínez, M. J.; Vazquez, G. E.; Díaz, M. F.; Ponzoni, I. Feature Learning Applied to the Estimation of Tensile Strength at Break in Polymeric Material Design. *J. Integr. Bioinformatics* **2016**, *13*, 286.

(29) Linati, L.; Lusvardi, G.; Malavasi, G.; Menabue, L.; Menziani, M. C.; Mustarelli, P.; Segre, U. Qualitative and Quantitative Structure-Property Relationships Analysis of Multicomponent Potential Bioglasses. *J. Phys. Chem. B* **2005**, *109*, 4989–4998.

(30) Lusvardi, G.; Malavasi, G.; Tarsitano, F.; Menabue, L.; Menziani, M. C.; Pedone, A. Quantitative Structure-Property Relationships of Potentially Bioactive Fluoro Phospho-Silicate Glasses. *J. Phys. Chem. B* **2009**, *113*, 10331–10338.

(31) Brauer, D. S. Bioactive Glasses - Structure and Properties. *Angew. Chem., Int. Ed.* **2015**, *54*, 4160–4181.

(32) Lusvardi, G.; Malavasi, G.; Menabue, L.; Menziani, M. C.; Pedone, A.; Segre, U. Density of Multicomponent Silica-Based Potential Bioglasses: Quantitative Structure-Property Relationships (QSPR) Analysis. *J. Eur. Ceram. Soc.* **2007**, *27*, 499–504.

(33) Lu, X.; Deng, L.; Gin, S.; Du, J. Quantitative Structure-Property Relationship (QSPR) Analysis of ZrO_2 -Containing Soda-Lime Borosilicate Glasses. *J. Phys. Chem. B* **2019**, *123*, 1412–1422.

(34) Lu, X.; Ren, M.; Deng, L.; Benmore, C. J.; Du, J. Structural Features of ISG Borosilicate Nuclear Waste Glasses Revealed from High-Energy X-Ray Diffraction and Molecular Dynamics Simulations. *J. Nucl. Mater.* **2019**, *515*, 284–293.

(35) Lu, X.; Du, J. Quantitative Structure-Property Relationship (QSPR) Analysis of Calcium Aluminosilicate Glasses Based on Molecular Dynamics Simulations. *J. Non-Cryst. Solids* **2020**, *530*, 119772.

(36) Plimpton, S. Fast Parallel Algorithms for Short-Range Molecular Dynamics. *J. Comput. Phys.* **1995**, *117*, 1–19.

(37) Deng, L.; Du, J. Development of Effective Empirical Potentials for Molecular Dynamics Simulations of the Structures and Properties of Boroaluminosilicate Glasses. *J. Non-Cryst. Solids* **2016**, *453*, 177–194.

(38) Deng, L.; Du, J. Development of Boron Oxide Potentials for Computer Simulations of Multicomponent Oxide Glasses. *J. Am. Ceram. Soc.* **2019**, *102*, 2482–2505.

(39) Deng, L.; Du, J. Development of Boron Oxide Potentials for Computer Simulations of Multicomponent Oxide Glasses. *J. Am. Ceram. Soc.* **2019**, *102*, 2482–2505.

(40) Thompson, A. P.; Plimpton, S. J.; Mattson, W. General Formulation of Pressure and Stress Tensor for Arbitrary Many-Body Interaction Potentials under Periodic Boundary Conditions. *J. Chem. Phys.* **2009**, *131*, 154107.

(41) Ren, M.; Deng, L.; Du, J. Bulk, Surface Structures and Properties of Sodium Borosilicate and Boroaluminosilicate Nuclear Waste Glasses from Molecular Dynamics Simulations. *J. Non-Cryst. Solids* **2017**, *476*, 87–94.

(42) Hill, R. The Elastic Behaviour of a Crystalline Aggregate. *Proc. Phys. Soc., London, Sect. A* **1952**, *65*, 349–354.

(43) Sun, K.-H. Fundamental condition of glass formation*. *J. Am. Ceram. Soc.* **1946**, *30*, 277–281.

(44) Tandia, A.; Vargheese, K. D.; Mauro, J. C.; Varshneya, A. K. Atomistic Understanding of the Network Dilation Anomaly in Ion-Exchanged Glass. *J. Non-Cryst. Solids* **2012**, *358*, 316–320.



CAS BIOFINDER DISCOVERY PLATFORM™

ELIMINATE DATA SILOS. FIND WHAT YOU NEED, WHEN YOU NEED IT.

A single platform for relevant, high-quality biological and toxicology research

Streamline your R&D

CAS
A division of the American Chemical Society